Technical Specification Sheet — Model XR-4500

The following specifications define the electrical, thermal, and mechanical operating parameters for the XR-4500 series precision voltage reference module. All values are valid at TA = +25 °C unless otherwise noted. Parameters marked with an asterisk (*) are guaranteed by design and characterization but not production-tested.

Parameter	Symbol	Min	Typ	Max	Unit	Tolerance	Test Condition	Temp	Notes
Input Voltage	VIN	4.5	5.0	5.5	V	±0.25 V	VOUT = 2.5 V, ILOAD = 0 mA	–40 to +85 °C	Regulated input
Output Voltage	VOUT	2.495	2.500	2.505	V	±0.1 %	VIN = 5.0 V, ILOAD = 10 mA	+25 °C	Trimmed at factory
Temp. Coefficient	TCVOUT	—	3	8	ppm/°C	±2.5 ppm	VIN = 5.0 V, box method	–40 to +125 °C	Grade A only
Load Regulation	ΔVOUT/ΔIL	—	0.5	2.0	mV/mA	—	0 mA ≤ ILOAD ≤ 50 mA	+25 °C	Sourcing only
Line Regulation	ΔVOUT/ΔVIN	—	0.1	1.0	mV/V	—	4.5 V ≤ VIN ≤ 15 V	+25 °C	DC measurement
Quiescent Current	IQ	—	800	1200	µA	±200 µA	ILOAD = 0 mA, VIN = 5.0 V	+25 °C	No-load condition
Output Noise (0.1–10 Hz)	eN_p-p	—	1.5	3.0	µVp-p	—	BW = 0.1 Hz to 10 Hz	+25 °C	Peak-to-peak
Power Supply Rejection	PSRR	80	100	—	dB	—	f = 120 Hz, VIN = 5.0 V ± 0.5 V	+25 °C	Ripple rejection
Thermal Resistance (J-A)	θJA	—	120	150	°C/W	—	Still air, JEDEC 2S2P board	—	SOIC-8 package